AMRITA PANDEY

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EDUCATION

Bachelor of Technology – Computer Science & Engineering *Aug 2023 – Present*
Lovely Professional University, Punjab, India | CGPA: 6.71

TECHNICAL SKILLS

Languages: Java, Python, C++, JavaScript, HTML, CSS, C
Frameworks & Libraries: Scikit-learn, OpenCV, NumPy, Pandas, SciPy, JavaAWT/Swing
Databases & Tools: MySQL (SQL & Relational DB), GitHub, Linux, VS Code, IntelliJ IDEA
CS Fundamentals: Object-Oriented Design, Data Structures & Algorithms, Complexity Analysis, Distributed Systems (conceptual)
Soft Skills: Problem-Solving, Effective Communication, Leadership, Self-Motivated, Fast Learner

PROJECTS

Nova – Personal Finance Tracker | JavaScript | HTML | CSS | Google OAuth | Netlify| LINK *Dec 2025-Jan 2026*

- Built and deployed a full-stack personal finance dashboard with real-time expense tracking, budget management, and spending analytics.
- Integrated Google OAuth authentication for secure, personalized user sessions, demonstrating ability to develop and deploy software integrated with cloud-based services.
- Designed a multi-section UI (Dashboard, Analytics, Budget) with dark mode, CSV export, and data visualization charts — showcasing strong web development and problem-solving skills.

Real-Time Heart Rate Estimation System | Python | OpenCV | Scikit-Learn | SciPy | NumPy| GitHub *Oct 2025 – Dec 2025*

- Engineered a real-time rPPG pipeline using OpenCV and YCrCb skin masking to isolate physiological signals from lighting and motion noise, demonstrating ability to build scalable, distributed signal processing systems.
- Implemented advanced signal processing algorithms (POS, CHROM, FFT) to extract high-fidelity pulse features from raw RGB video — showcasing complex problem-solving in software systems.
- Architected a Weighted Voting Ensemble (Random Forest, GBM, Extra Trees) optimized via Grid Search, achieving ~90% accuracy (MAE < 5 BPM) through automated model evaluation pipelines.

Bank Management System | Java | AWT | Swing | OOP | Event-Driven Programming| GitHub *Jun 2025 – Jul 2025*

- Constructed a scalable desktop application using Java Swing with a modular, event-driven architecture — directly applying object-oriented design principles and separation of concerns.
- Designed a multi-panel dashboard with dynamic financial data (balance, FDs, CIBIL score) and quick-action utilities (transfers, payments, calculators), demonstrating UI development skills.
- Implemented robust screen transitions and responsive UI, showcasing ability to develop and deploy software of exceptional quality in a structured development environment.

TRAINING

Java Maestro: Hands-On GUI Application Development | Lovely Professional University *Jun 2025 – Jul 2025*

- Built a functional Bank Management System during structured training, implementing core banking operations (account creation, deposits, withdrawals, balance management) using Java AWT/Swing.
- Applied event-driven programming, modular coding practices, and debugging techniques through daily guided sessions — strengthening code architecture and real-world software delivery skills.

ACHIEVEMENTS & CERTIFICATIONS

- Solved 200+ coding problems across platforms, demonstrating strong consistency in algorithmic problem-solving.
- Attended Agentic AI Day powered by Google (Sep 2025) — staying current with emerging cloud-based AI technologies.
- Certifications: Data Structures using C++ | Microlearning in Full Stack Development | Deloitte Cyber Job Simulation